



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

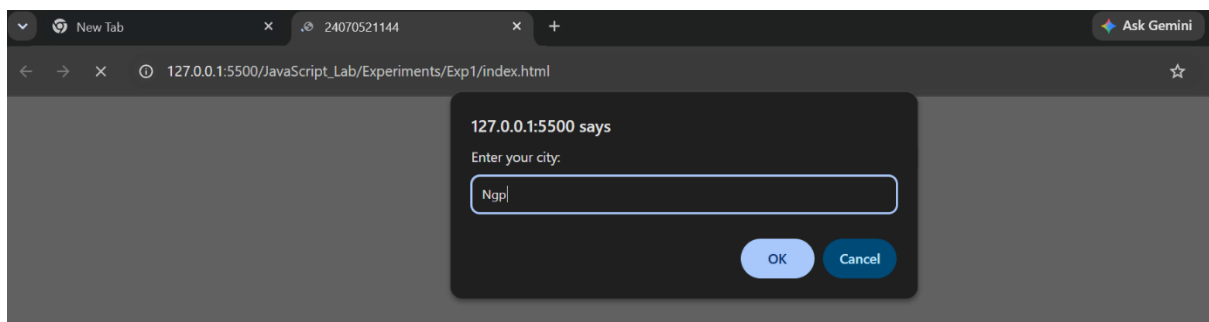
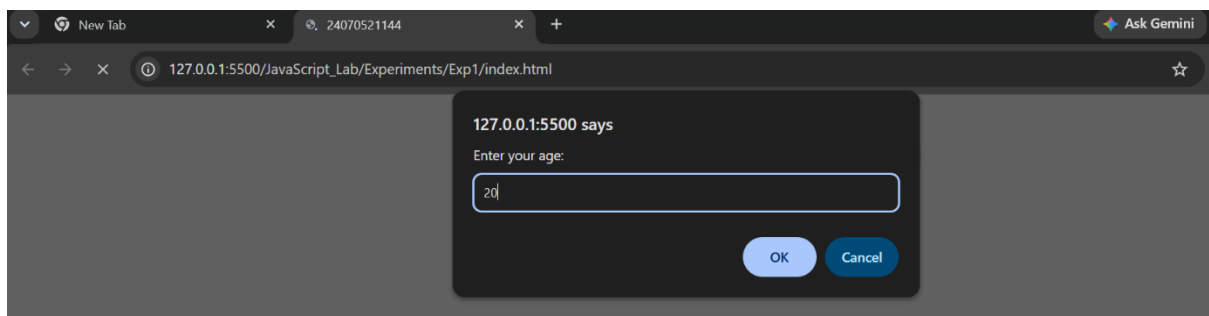
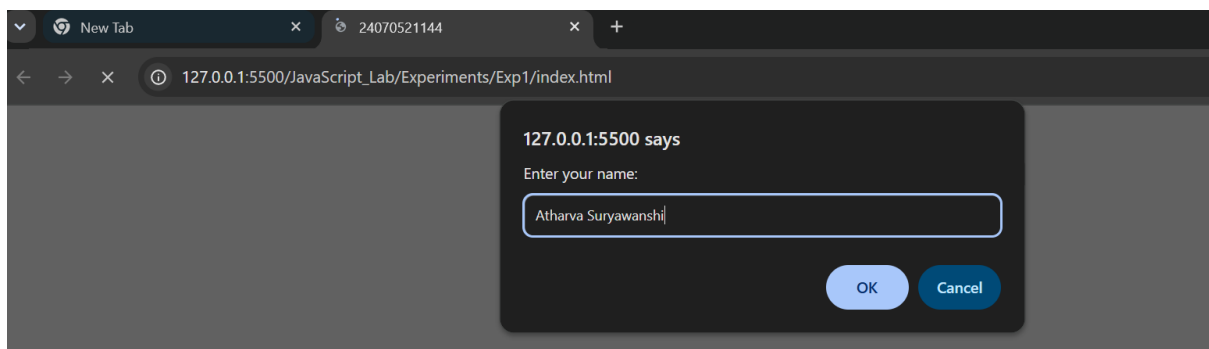
Experiment 01

Aim: Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Theory: JavaScript can be included in a web page using inline, internal, and external methods. Inline JavaScript is written directly within HTML elements, internal JavaScript is placed inside the

Output:





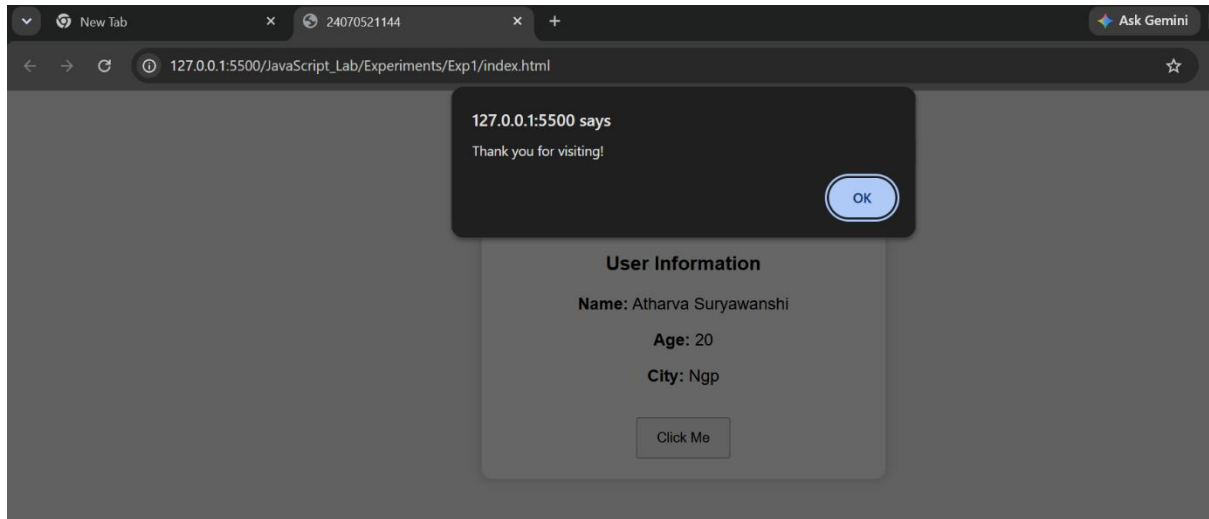
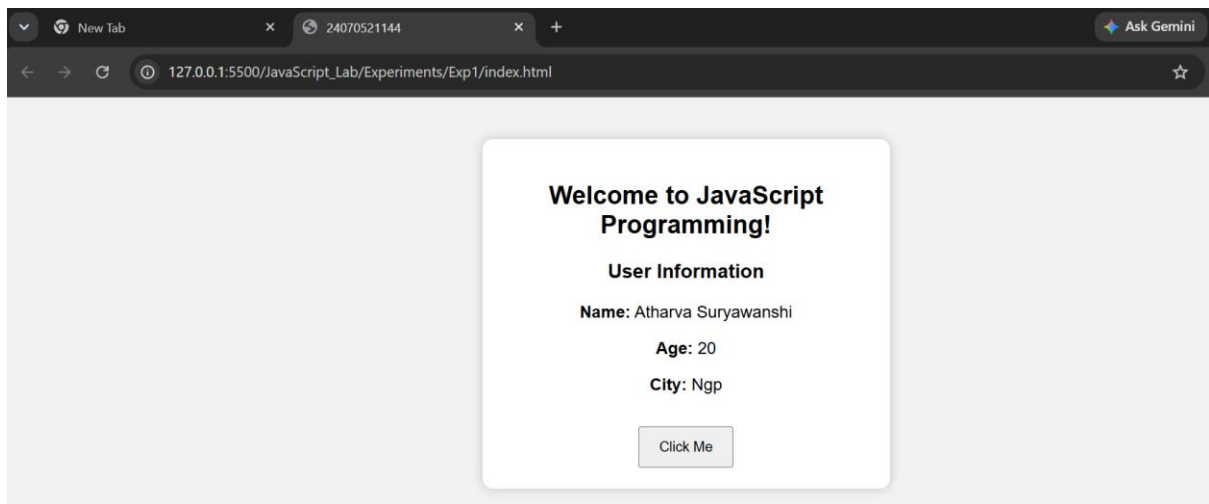
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Result: The webpage was successfully created using inline, internal, and external JavaScript. It displayed a welcome message, accepted user information through input prompts, and demonstrated various console methods such as `console.log()`, `console.info()`, `console.warn()`, `console.error()`, and `console.table()` for debugging and output.



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Case Study 01

Aim: Create a webpage for college department and link the student information page with it and write a javascript program to display “Welcome to SIT Nagpur” in an alert box.

Program Code:

```
Case_Study > Case_Study1 > student_info.html > ...
You, 2 weeks ago | 1 author (You)
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Student Information</title>
7 <style> ...
46 </style>
47 </head>
48 <body>
49
50 <h1>Student Information Directory</h1>
51 <p>Below is the list of current students enrolled in the department.</p>
52
53 <table>
54 <thead>
55 <tr>
56 <th>Roll Number</th>
57 <th>Name</th>
58 <th>Year</th>
59 <th>Email</th>
60 </tr>
61 </thead>
62 <tbody>
63 <tr>
64 <td>CS101</td>
65 <td>Alice Smith</td>
66 <td>3rd Year</td>
67 <td>alice@college.edu</td>
68 </tr>
69 <tr>
70 <td>CS102</td>
71 <td>Bob Jones</td>
72 <td>2nd Year</td>
73 <td>bob@college.edu</td>
74 </tr>
75 <tr>
76 <td>CS103</td>
77 <td>Charlie Brown</td>
78 <td>4th Year</td>
79 <td>charlie@college.edu</td>
80 </tr>
81 </tbody>
82 </table>
83
84 <a href="department.html" class="back-btn">Back to Department Home</a>
85
86 </body>
87 </html>
```



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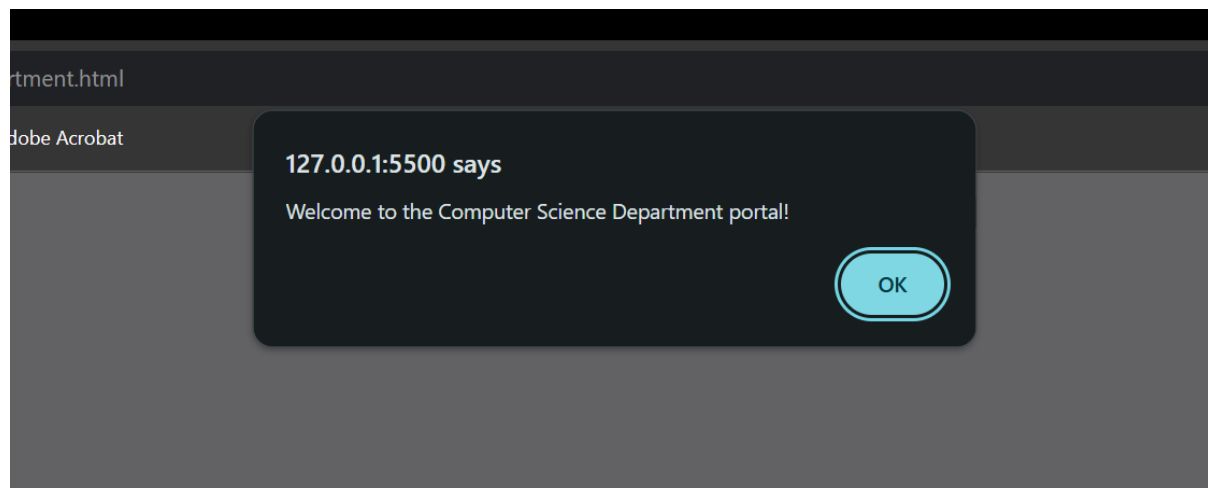
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```
...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>Computer Science Department</title>
7  >
52 </style>
53 </head>
54 <body>
55
56 <header>
57   <h1>Symbiosis Institute of Technology</h1>
58   <p>Wathoda Nagpur</p>
59 </header>
60
61 <nav>
62   <a href="#">Home</a>
63   <a href="student_info.html">Student Information</a>
64   <a href="#about">About Us</a>
65 </nav>
66
67 <main>
68   <section id="about">
69     <h2>About the Department</h2>
70     <p>Welcome to the Computer Science Department.
71       We offer cutting-edge programs designed to prepare students for the rapidly evolving tech industry.
72       Our curriculum covers software engineering, artificial intelligence, data science, and cybersecurity.
73     </p>
74   </section>
75 </main>
76
77 <footer>
78   <p>&copy; 2026 College Name. All rights reserved.</p>
79 </footer>
80
81 <script>
82   window.onload = function() {
83     alert("Welcome to the Computer Science Department portal!");
84   };
85 </script>
86
87 </body>
88 </html>
```

Output:





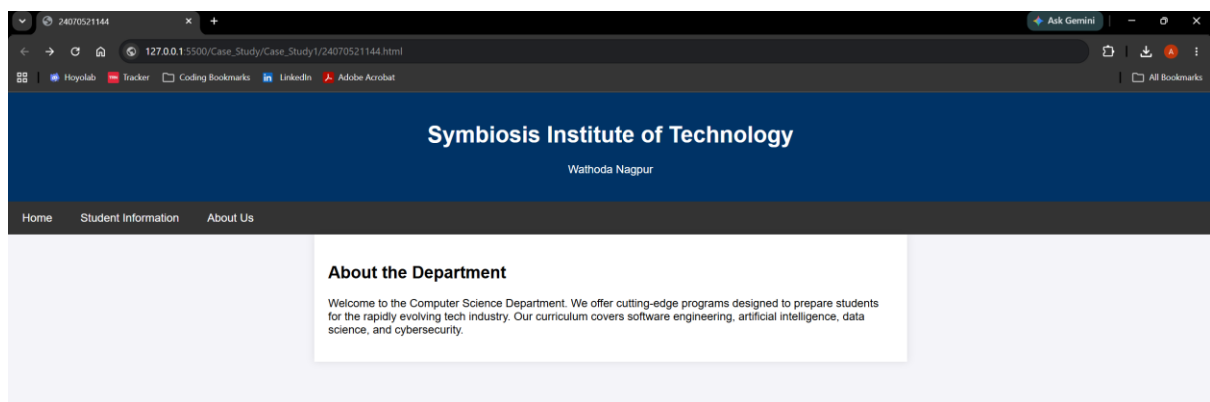
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Result: A webpage for the college department was successfully created using HTML, CSS, and JavaScript. Student information was linked and displayed correctly, and the JavaScript program successfully displayed the message **"Welcome to SIT Nagpur"** on the webpage. Thus, the aim of creating a college department webpage and implementing JavaScript functionality was achieved successfully.



Experiment: 02

Aim: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Theory: JavaScript provides var, let, and const for declaring variables with different scopes and behaviors. It also supports modern ES6 features such as template literals for easy string formatting and object destructuring for extracting object properties into variables. The prompt() function is used to accept input from the user. In this experiment, these features are used to develop a billing calculator that accepts product details, calculates the subtotal, GST (18%), and total amount, and displays a formatted bill dynamically on the web page.

127.0.0.1:5500 says
Enter Customer Name:
User
OK Cancel

127.0.0.1:5500 says
Enter Product Name:
Coffee
OK Cancel

127.0.0.1:5500 says
Enter Product Price (₹):
249
OK Cancel

127.0.0.1:5500 says
Enter Quantity:

OK Cancel



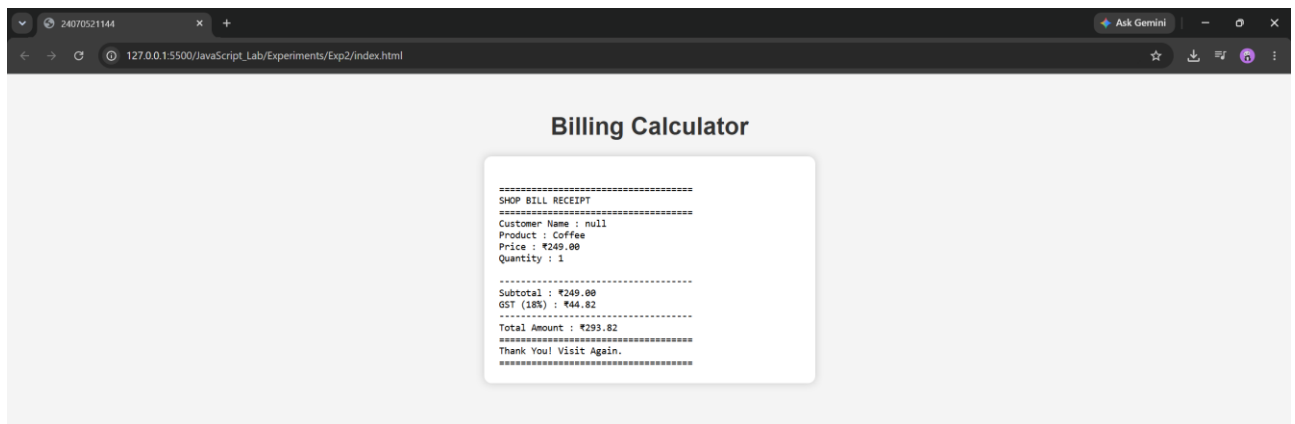
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Result : The billing calculator was successfully developed using JavaScript. It accepted user input for the customer name, product details, price, and quantity, calculated the subtotal, GST (18%), and total amount, and displayed a well-formatted billing receipt on the web page.



Case Study 02

Aim: Develop the webpage for an inline shopping website with using html, css and html tags and calculate the bill, customer name, mobile number, invoice number, product number, quantity(kg), rate per kg, discount, GST, packing charges, payment mode(cash/card/UPI), membership(yes/no)

Program Code:

```
1 <!DOCTYPE html> You, last week + Add files via upload
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>24070521144</title>
7   <!-- Link to external CSS file -->
8   <link rel="stylesheet" href="style.css">
9 </head>
10 <body>
11
12 <div class="container">
13   <h2>Online Shopping Checkout & Bill</h2>
14   <form id="billForm">
15     <div class="form-group">
16       <label for="custName">Customer Name:</label>
17       <input type="text" id="custName" required placeholder="Enter name">
18     </div>
19     <div class="form-group">
20       <label for="mobNo">Mobile Number:</label>
21       <input type="tel" id="mobNo" required placeholder="10-digit mobile number">
22     </div>
23     <div class="form-group">
24       <label for="invNo">Invoice Number:</label>
25       <input type="text" id="invNo" required value="INV-2026-001">
26     </div>
27     <div class="form-group">
28       <label for="prodName">Product Name:</label>
29       <input type="text" id="prodName" required placeholder="e.g., Organic Sugar">
30     </div>
31     <div class="form-group">
32       <label for="qtyKg">Quantity (in kg):</label>
33       <input type="number" id="qtyKg" step="0.1" min="0.1" required value="1">
34     </div>
35     <div class="form-group">
36       <label for="rateKg">Rate per kg (₹):</label>
37       <input type="number" id="rateKg" step="1" min="0" required value="100">
38     </div>
39     <div class="form-group">
40       <label for="discountPct">Discount (%):</label>
41       <input type="number" id="discountPct" step="0.1" min="0" max="100" value="5">
42     </div>
43     <div class="form-group">
44       <label for="gstPct">GST (%):</label>
45       <input type="number" id="gstPct" step="0.1" min="0" max="100" value="5">
46     </div>
47     <div class="form-group">
48       <label for="packingCharges">Packing Charges (₹):</label>
49       <input type="number" id="packingCharges" step="1" min="0" value="20">
50     </div>
51   </form>
52 </div>
```

it Graph

🕒 You, last week 🕒 AtharvaSuryas (1 week ago)



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```
45 <input type="number" id="gstPct" step="0.1" min="0" max="100" value="5">
46 </div>
47 <div class="form-group">
48 <label for="packingCharges">Packing Charges (₹):</label>
49 <input type="number" id="packingCharges" step="1" min="0" value="20">
50 </div>
51 <div class="form-group">
52 <label for="membership">Membership Type:</label>
53 <select id="membership">
54 <option value="none">None (0% extra off)</option>
55 <option value="silver">Silver (2% extra off)</option>
56 <option value="gold">Gold (5% extra off)</option>
57 </select>
58 </div>
59 <div class="form-group">
60 <label for="paymentMode">Payment Mode:</label>
61 <select id="paymentMode">
62 <option value="UPI">UPI</option>
63 <option value="Card">Credit/Debit Card</option>
64 <option value="COD">Cash on Delivery</option>
65 </select>
66 </div>
67 <button type="button" id="calcBtn">Calculate Bill</button>
68 </form>
69
70 <div class="bill-summary" id="billSummary">
71 <h3>Invoice Details</h3>
72 <div class="bill-row"><span>Customer:</span> <span id="outName"></span></div>
73 <div class="bill-row"><span>Mobile:</span> <span id="outMob"></span></div>
74 <div class="bill-row"><span>Invoice ID:</span> <span id="outInv"></span></div>
75 <div class="bill-row"><span>Product:</span> <span id="outProd"></span></div>
76 <div class="bill-row"><span>Gross Price:</span> <span id="outGross"></span></div>
77 <div class="bill-row"><span>Discount Applied:</span> <span id="outDisc"></span></div>
78 <div class="bill-row"><span>Packing Charges:</span> <span id="outPack"></span></div>
79 <div class="bill-row"><span>GST Amount:</span> <span id="outGst"></span></div>
80 <div class="bill-row"><span>Payment & Membership:</span> <span id="outMeta"></span></div>
81 ... <div class="bill-row total-row"><span>Final Payable:</span> <span id="outTotal"></span></div>
82 </div>
83 </div>
84
85 <!-- Link to external JS file -->
86 <script src="script.js"></script>
87 </body>
88 </html>
```

Git Graph

🕒 You, last week 🕒 AtharvaSuryas (1 week ago)



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Output:

The screenshot shows a web browser window with the URL `127.0.0.1:5500/Case_Study/Case%20Study%202/index.html`. The page is titled "Online Shopping Checkout & Bill". It contains a form with the following fields:

- Customer Name: User
- Mobile Number: 1234567890
- Invoice Number: INV-2026-001
- Product Name: Pudina
- Quantity (in kg): 1
- Rate per kg (₹): 100
- Discount (%): 5
- GST (%): 5
- Packing Charges (₹): 20
- Membership Type: Silver (2% extra off)
- Payment Mode: Cash on Delivery

A green button labeled "Calculate Bill" is at the bottom of the form.

This screenshot shows the same form as the previous one, but with the "Calculate Bill" button clicked. Below the form, an "Invoice Details" section is displayed, showing the calculated values:

Customer:	User
Mobile:	1234567890
Invoice ID:	INV-2026-001
Product:	Pudina (1 kg @ ₹100/kg)
Gross Price:	₹100.00
Discount Applied:	-₹7.00 (7%)
Packing Charges:	₹20.00
GST Amount:	+₹4.65 (5%)
Payment & Membership:	COD SILVER Member
Final Payable:	₹117.65



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Result: The online shopping billing webpage was successfully developed using HTML, CSS, and JavaScript. The program accepted customer and product details, calculated the total bill including discount, GST, and packing charges, and displayed the final invoice correctly. It demonstrated the use of user input, arithmetic calculations, objects, template literals, and object destructuring in JavaScript.



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Experiment No: 03

Aim: Implement control structures and form validation, create a grading system based on user entered marks

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Theory: JavaScript provides essential control structures, including conditional statements (if, else if, else, and switch) and loops (for, while, and do-while), to manage program execution based on dynamic conditions. These structures enable developers to build interactive features such as form validation, which checks and sanitizes user input on the client side before submission to prevent invalid or incomplete data. Additionally, JavaScript interacts with DOM elements to extract input values, process business logic—such as evaluating numeric ranges to assign grades—and dynamically display results on the web page. In this experiment, control structures and form validation techniques are applied to capture student marks, validate the input, dynamically calculate a grade, and display the formatted outcome to the user.

Output:

Student Grading System

Enter Marks (0 - 100):

Calculate Grade

Your Grade is: A+



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Student Grading System

Enter Marks (0 - 100):

101

Calculate Grade

Error: Please enter a valid number between 0 and 100.

Result:

The JavaScript-based control structures and form validation system were successfully implemented and tested. The application correctly validates user-entered marks, ensuring input fields are non-empty, numeric, and fall within the valid range of 0 to 100. Upon validation, the control logic evaluates the entered score against predefined grading criteria and dynamically displays the corresponding grade along with feedback directly on the web page.



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Case Study 03

Aim: Scholarship Eligibility, A college awards scholarships based on marks. Below 50-Not Eligible, 50-69-Bronze, 70-84-Silver, 85-94-Gold, 95-100-Platinum.

Program Code:

```
index.html ...\Case Study 3 U X index.html ...\Case Study 2 M
Case_Study > Case Study 3 > index.html > html > title
1 <!DOCTYPE html>
2 <html>
3   <title>
4     24070521144
5   </title>
6 <head>
7   <style>
8     body{font-family:Arial,sans-serif;background: #f0f2f5;margin:0;padding:20px}
9     .card{background: #fff;padding:30px;border-radius:10px;box-shadow:0 4px 10px rgba(0,0,0,0.1);text-align:center;width:260px}
10    h2{margin-top:0;color: #333}
11    input{padding:8px;width:80%;border:1px solid #ccc;border-radius:5px;margin-bottom:12px}
12    button{padding:8px 16px;border:none;background: #4CAF50;color: #fff;border-radius:5px;cursor:pointer}
13    button:hover{background: #45a049}
14    #r{margin-top:15px;font-weight:bold;font-size:18px;color: #222}
15  </style>
16 </head>
17 <body>
18   <div class="card">
19     <h2>Scholarship Eligibility</h2>
20     Marks: <input type="number" id="m">
21     <button onclick="check()">Check</button>
22     <p id="r"></p>
23   </div>
24
25   <script>
26   function check(){
27     let m = Number(document.getElementById('m').value);
28     let r;
29     if(m<50) r="Not Eligible";
30     else if(m<70) r="Bronze";
31     else if(m<85) r="Silver";
32     else if(m<95) r="Gold";
33     else r="Platinum";
34     document.getElementById('r').innerText = "Result: " + r;
35   }
36 </script>
37 </body>
38 </html>
```



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Output:

Result: The Scholarship Eligibility was successfully implemented using JavaScript. The program validated the user input, assigned eligibility marks using if-else control structures, and displayed the correct Eligibility awards based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.



Experiment 04

Aim: Use function types, scope and closures; apply try-catch; build a palindrome checker

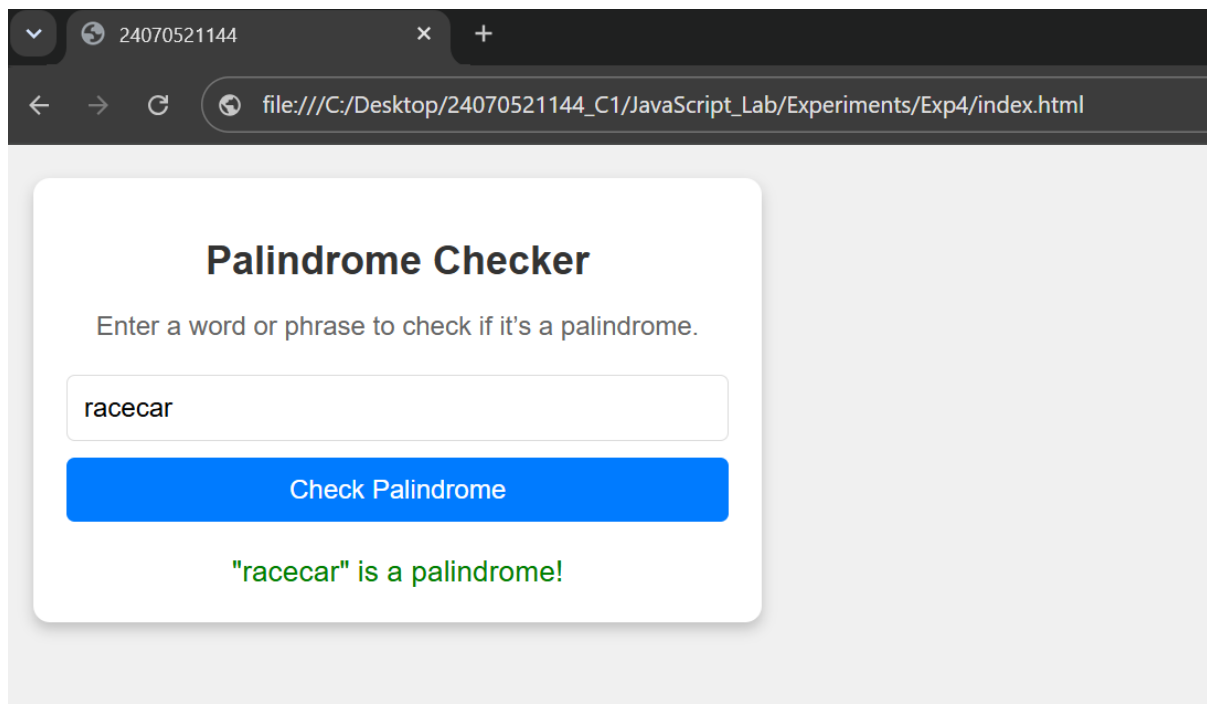
Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Theory: A palindrome is a word, number, or sentence that reads the same forward and backward, such as MADAM, LEVEL, and 121. A palindrome checker is a program that checks whether the given input is a palindrome or not by comparing the original string with its reverse.

In JavaScript, different types of functions, such as function declarations, function expressions, and arrow functions, can be used to implement the palindrome checker. Variable scope determines where variables can be accessed in the program, while closures allow an inner function to access variables from its outer function.

The try-catch statement is used for exception handling and helps manage errors, such as invalid or empty input, without stopping the execution of the program. Thus, a palindrome checker demonstrates the concepts of functions, scope, closures, and error handling in JavaScript.

Output:





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Palindrome Checker

Enter a word or phrase to check if it's a palindrome.

burger

Check Palindrome

"burger" is not a palindrome.

Result: Successfully implemented a Palindrome detector using JavaScript ES6 features, including var, let, const, template literals, and destructuring, to find and display whether the given text or number is a palindrome or not.



Case Study 04

Aim: The ATM system verifies if the pin is palindrome. If the palindrome displays a security message. To implement this create a function to reverse the entered pin number and another function to check if it is palindrome. use function declaration, function expression, arrow function and demonstrate scope and closure concepts.

Program Code:

```
Case_Study > Case Study 4 > index.html > ...
You, 2 days ago | 1 author (You)
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>PIN Strength Checker</title>
7   <style>...
58 </style>
59 </head>
60 <body>
61
62 <div class="container">
63   <h2>PIN Strength Checker</h2>
64   <p>Enter your PIN below to check its strength.</p>
65   <input type="text" id="pinInput" placeholder="Enter PIN" maxlength="8">
66   <br>
67   <button onclick="checkPinStrength()">Check Strength</button>
68   <div id="result"></div>
69 </div>
70
71 <script>
72 function checkPinStrength() {
73   const pinInput = document.getElementById('pinInput').value;
74   const resultDiv = document.getElementById('result');
75
76   // Clear previous results and check for empty input
77   resultDiv.innerHTML = '';
78   if (!pinInput) {
79     resultDiv.innerHTML = 'Please enter a PIN.';
80     resultDiv.className = '';
81     return;
82   }
83
84   // Reverse the string to check for palindrome
85   const reversedPin = pinInput.split('').reverse().join('');
86
87   if (pinInput === reversedPin) {
88     resultDiv.innerHTML = '⚠ Weak PIN (Palindrome)';
89     resultDiv.className = 'weak';
90   } else {
91     resultDiv.innerHTML = '✅ Strong PIN';
92     resultDiv.className = 'strong';
93   }
94 }
95 </script>
96
97 </body>
98 </html>
```



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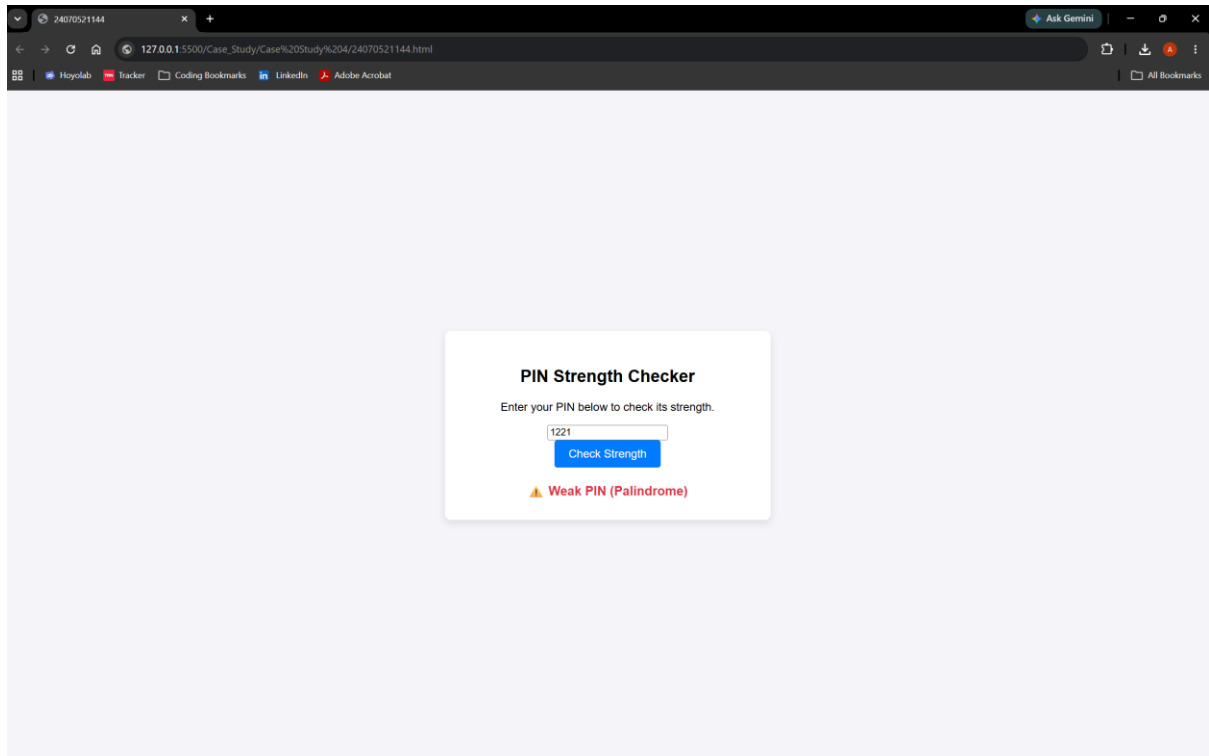
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Output:



Result: Successfully implemented a ATM pin Palindrome detector using JavaScript ES6 features, including var, let, const, template literals, and destructuring, to find and display whether the given ATM pin is a palindrome or not.

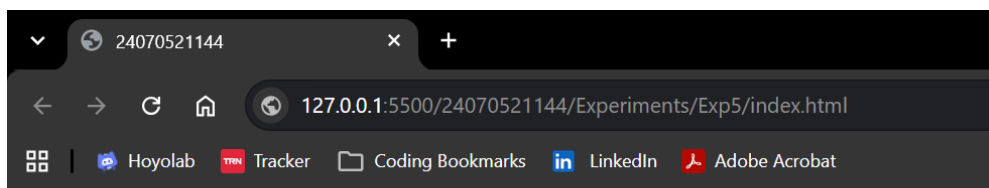


Experiment 05

Aim: Apply array methods and object handling, create a cart total calculator with discount logic

Theory: JavaScript provides high-order array methods and object manipulation capabilities to model real-world data structures efficiently. Objects allow grouping related key-value pairs—such as product names, prices, and quantities—into structured data models. Native JavaScript array methods like `.map()` transform dataset arrays into dynamic visual representations (such as HTML table rows), `.filter()` creates new arrays by excluding specific elements (enabling item deletion), and `.reduce()` aggregates multi-item values into a single total amount. Additionally, control logic can be chained with these calculations to dynamically evaluate threshold conditions—such as applying tiered percentage discounts based on total order values. In this experiment, these concepts are combined to build a shopping cart calculator that processes item object arrays, evaluates tier-based discount rules, and updates the final payable amount on the DOM in real time.

Output:



Shopping Cart

Item	Price (₹)	Qty	Subtotal (₹)	Action
Headphones	₹1200.00	1	₹1200.00	<button>Remove</button>
Mouse	₹450.00	2	₹900.00	<button>Remove</button>
USB Cable	₹200.00	3	₹600.00	<button>Remove</button>

Subtotal: ₹2700.00

Discount Tier: 20%

Discount Applied: -₹540.00

Final Total: ₹2160.00



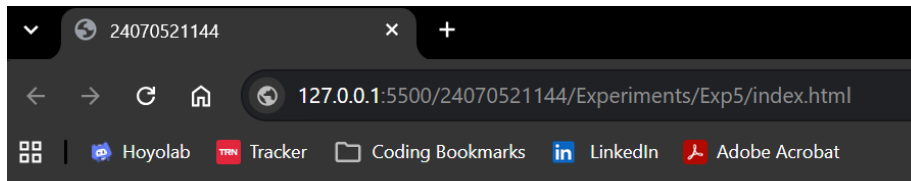
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Shopping Cart

Item	Price (₹)	Qty	Subtotal (₹)	Action
Headphones	₹1200.00	1	₹1200.00	<button>Remove</button>
Mouse	₹450.00	3	₹1350.00	<button>Remove</button>
USB Cable	₹200.00	3	₹600.00	<button>Remove</button>

Subtotal: ₹3150.00

Discount Tier: 30%

Discount Applied: -₹945.00

Final Total: ₹2205.00

Result:

The shopping cart total calculator was successfully implemented using JavaScript array methods (.map(), .filter(), .reduce()) and object handling. The application accurately computes cart subtotals, handles item removal, and applies slab discounts (10% at ₹1,000, 20% at ₹2,000, and 30% at ₹3,000+) to update the final total in real time.

Conclusion:

This experiment highlights how array methods and object-oriented data structures simplify data manipulation and aggregation in JavaScript. Combining these methods with conditional discount logic enables efficient client-side calculations and dynamic DOM updates for interactive web applications.



Case Study 05

Aim: Study and Implementation of Arrays and Loops in JavaScript

Program Code:

```
JavaScript_Lab > Case_Study > Case Study 5 > index.html > html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>Interactive Arrays & Loops</title>
7    <!-- Google Font for modern typography -->
8    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap" rel="stylesheet">
9
10 > <style>...
335 </style>
336 </head>
337 <body>
338   <header>
339     <div class="header-badge">JS LAB EXPERIMENT</div>
340     <h1>Interactive Arrays & Loops</h1>
341     <p>Explore JavaScript Array Methods in Real Time</p>
342   </header>
343
344   <main class="container">
345     <!-- Section 1 -->
346     <section class="section">
347       <h2>1. Interactive Array</h2>
348       <p class="section-desc">
349         Add numbers and experiment with different JavaScript array manipulation methods.
350       </p>
351
352       <div id="mainArray" class="array-display"></div>
353
354       <div class="control-row">
355         <input
356           type="number"
357           id="numberInput"
358           placeholder="Enter a number"
359         >
360         <button onclick="addNumber()" class="btn btn-primary">
361           + Add Number
362         </button>
363       </div>
364
365       <div class="button-group">
366         <button onclick="removeLast()" class="btn btn-danger">
367           pop()
368         </button>
369         <button onclick="removeFirst()" class="btn btn-danger">
370           shift()
371         </button>
372         <button onclick="addBeginning()" class="btn btn-success">
373           unshift()
374         </button>
375         <button onclick="clearArray()" class="btn btn-outline-danger">
376           Clear Array
377         </button>
378       </div>
379     </section>
380
381     <!-- Section 2 -->
382     <section class="section">
383       <h2>2. Array Manipulation Methods</h2>
384
```



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```
382 <section class="section">
383   <h2>2. Array Manipulation Methods</h2>
384
385   <div class="grid-2col">
386     <div class="operation-box">
387       <h3>splice()</h3>
388       <p>Replace an element at a selected position.</p>
389       <div class="input-stack">
390         <input
391           type="number"
392           id="spliceIndex"
393           placeholder="Index"
394           min="0"
395         >
396         <input
397           type="number"
398           id="spliceValue"
399           placeholder="New Value"
400         >
401         <button onclick="spliceArray()" class="btn btn-primary btn-full">
402           Apply splice()
403         </button>
404       </div>
405     </div>
406
407     <div class="operation-box">
408       <h3>slice()</h3>
409       <p>Extract a portion without modifying original array.</p>
410       <div class="input-stack">
411         <input
412           type="number"
413           id="sliceStart"
414           placeholder="Start"
415           min="0"
416         >
417         <input
418           type="number"
419           id="sliceEnd"
420           placeholder="End"
421           min="0"
422         >
423         <button onclick="sliceArray()" class="btn btn-primary btn-full">
424           Apply slice()
425         </button>
426       </div>
427     </div>
428   </div>
429
430   <div class="output-wrapper">
431     <span class="output-label">Console Output</span>
432     <div id="manipulationOutput" class="output">
433       Select an operation to see its result.
434     </div>
435   </div>
436 </section>
437
438 <!-- Section 3 -->
```



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```
439 <section class="section">
440 <h2>3. Array Iteration Methods</h2>
441 <p class="section-desc">
442 | Choose an operation to process the current array.
443 </p>
444
445 <div class="method-buttons">
446 | <button onclick="runMap()" class="btn btn-accent">
447 | | map()
448 | </button>
449 | <button onclick="runFilter()" class="btn btn-accent">
450 | | filter()
451 | </button>
452 | <button onclick="runReduce()" class="btn btn-accent">
453 | | reduce()
454 | </button>
455 | <button onclick="runForEach()" class="btn btn-accent">
456 | | forEach()
457 | </button>
458 </div>
459
460 <div class="output-wrapper">
461 | <span class="output-label">Iteration Result</span>
462 | <div id="iterationOutput" class="output">
463 | | Click a method to see how it works.
464 | </div>
465 </div>
466 </section>
467
468 <!-- Section 4 -->
469 <section class="section case-study">
470 <h2>4. Case Study: Maximum & Minimum</h2>
471 <p class="section-desc">
472 | Find the maximum and minimum values from the target dataset:
473 </p>
474
475 <div class="array-display static-array">
476 | [25, 10, 45, 5, 30, 15]
477 </div>
478
479 <button onclick="runCaseStudy()" class="btn btn-case-study">
480 | 🔍 Find Maximum & Minimum
481 </button>
482
483 <div class="result-grid">
484 | <div class="result-card max-card">
485 | | <h3>Maximum Value</h3>
486 | | <strong id="maximum">-</strong>
487 | </div>
488 | <div class="result-card min-card">
489 | | <h3>Minimum Value</h3>
490 | | <strong id="minimum">-</strong>
491 | </div>
492 </div>
493
494 <div class="output-wrapper">
495 | <span class="output-label">Execution Details</span>
```



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```
652 console.log("Reduce():", sum);
653 document.getElementById("iterationOutput").textContent =
654     "reduce() → Calculates the sum\n\n" +
655     "Array:\n[" +
656     numbers.join(", ") +
657     "]\n\n" +
658     "Sum = " +
659     sum;
660 }
661
662 function runForEach() {
663     let result = "";
664     numbers.forEach(function(value, index) {
665         result += "Index " + index + " → " + value + "\n";
666         console.log("forEach():", index, value);
667     });
668     document.getElementById("iterationOutput").textContent =
669         "forEach() → Iterates through every element\n\n" + result;
670 }
671
672 function clearArray() {
673     numbers = [];
674     console.log("Array cleared:", numbers);
675     displayArray();
676     document.getElementById("manipulationOutput").textContent =
677         "Array cleared successfully.";
678 }
679
680 function runCaseStudy() {
681     let caseStudyArray = [25, 10, 45, 5, 30, 15];
682     let maximum = caseStudyArray.reduce(function(max, value) {
683         return value > max ? value : max;
684     });
685     let minimum = caseStudyArray.reduce(function(min, value) {
686         return value < min ? value : min;
687     });
688     document.getElementById("maximum").textContent = maximum;
689     document.getElementById("minimum").textContent = minimum;
690     document.getElementById("caseStudyOutput").textContent =
691         "Array: [" +
692         caseStudyArray.join(", ") +
693         "]\n\n" +
694         "Using reduce() to compare each element:\n\n" +
695         "Maximum Value: " +
696         maximum +
697         "\n" +
698         "Minimum Value: " +
699         minimum;
700     console.log("Case Study Array:", caseStudyArray);
701     console.log("Maximum Value:", maximum);
702     console.log("Minimum Value:", minimum);
703 }
704
705 // Initial render
706 displayArray();
707 </script>
708 </body>
709 </html>
```



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Output:

Interactive Arrays & Loops

Ask Gemini

127.0.0.1:5500/JavaScript_Lab/Case_Study/Case%20Study%205/index.html

Hoyolab Tracker Coding Bookmarks LinkedIn Adobe Acrobat All Bookmarks

JS LAB EXPERIMENT

Interactive Arrays & Loops
Explore JavaScript Array Methods in Real Time

1. Interactive Array

Add numbers and experiment with different JavaScript array manipulation methods.

[25, 10, 45, 5, 30, 15]

Enter a number

+ Add Number

pop() shift() unshift() Clear Array

2. Array Manipulation Methods

splice()
Replace an element at a selected position.

Index

New Value

Apply splice()

slice()
Extract a portion without modifying original array.

Start

End

Apply slice()

CONSOLE OUTPUT
Select an operation to see its result.

3. Array Iteration Methods

Choose an operation to process the current array.

map() filter() reduce() forEach()

ITERATION RESULT
Click a method to see how it works.



3. Array Iteration Methods

Choose an operation to process the current array.

map()

filter()

reduce()

forEach()

ITERATION RESULT

Click a method to see how it works.

4. Case Study: Maximum & Minimum

Find the maximum and minimum values from the target dataset:

[25, 10, 45, 5, 30, 15]

Find Maximum & Minimum

MAXIMUM VALUE

—

MINIMUM VALUE

—

EXECUTION DETAILS

Click the button to calculate the result.

Result: The interactive array and loop demonstration system was successfully implemented and tested. The web interface correctly performs real-time array mutations (push(), pop(), shift(), unshift(), splice(), and slice()) and updates the displayed array dynamically. High-order iteration methods (map(), filter(), reduce(), and forEach()) execute correctly and present their transformed or aggregated outputs in the output console blocks. Additionally, the case study algorithm successfully evaluated the dataset [25, 10, 45, 5, 30, 15] using the .reduce() method, accurately returning **45** as the maximum value and **5** as the minimum value.



Experiment No 6

Aim: Use string function and regex for email validation, data extraction and text analysis

Theory: This practical demonstrates basic web development and text manipulation using HTML, CSS, and JavaScript. The user interface takes string inputs and processes them dynamically in real time using DOM manipulation and standard string methods. Regular Expressions (Regex) are used to search, pattern-match, and extract valid email addresses from unformatted text blocks. Built-in string methods like `.toUpperCase()`, `.toLowerCase()`, and `.length` enable instant case conversion and character counting. Word counts and targeted keyword tracking (e.g., detecting "welcome" and counting "email") are performed by splitting strings and using pattern matching (`RegExp.prototype.test()` and `String.prototype.match()`). Overall, the project demonstrates client-side data validation, text processing, and dynamic UI rendering without requiring page reloads.

Outputs:

The screenshot shows a web browser window with the URL `C:/Users/ASUS/OneDrive/Desktop/24070521144_C1/JavaScript_Lab/Experiments/Exp6/index.html`. The application is titled "Text & Email Analyzer". It features a text input area with the placeholder text: "Hello and welcome to this website. Please enter you email To contact us reach to help@gmail.com or support@mail.com". Below the input area are two buttons: "UPPERCASE" and "lowercase". Underneath these buttons are three boxes displaying statistics: "115 Characters", "18 Words", and "1 'email' Count". The "Analysis Results" section shows: "Contains 'welcome': Yes" (with a green checkmark), "Extracted Emails (Regex):" followed by a list of emails: "help@gmail.com" and "support@mail.com", and "Single Email Verification:" followed by a red box indicating "Invalid / Not a single email".



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Result:

The developed web application successfully processes user input in real time. It accurately measures total character length and word count, detects the presence of the word "welcome", and counts occurrences of the word "email". Furthermore, the uppercase and lowercase transformation buttons dynamically reformat the text, while the Regular Expression engine correctly extracts all valid email addresses embedded in the text block and validates individual email formats.

Conclusion:

In conclusion, this practical successfully demonstrates the core concepts of client-side web development using HTML, CSS, and JavaScript. By integrating Regular Expressions with DOM manipulation and built-in JavaScript string functions, the application achieves dynamic, real-time text analysis and email validation. This highlights how front-end scripts can manipulate and validate user data instantly without requiring backend server processing or web page reloads.



Case Study 06

Aim: Study and Implementation of String Operations, String Manipulation Methods, and Vowel Counting in JavaScript

Program Code:

```
JavaScript_Lab > Case_Study > Case Study 6 > index.html > ...
You, 27 minutes ago | author (You)
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>JavaScript String Operations Lab</title>
7   <style>...
80 </style>
81 </head>
82 <body>
83
84 <div class="container">
85   <h1>JavaScript String Operations</h1>
86
87   <!-- Part A & B: Basics & Methods -->
88   <div class="section">
89     <h2>a & b) Basics and Built-in Methods</h2>
90     <div class="form-group">
91       <label for="mainString">Input String:</label>
92       <input type="text" id="mainString" value="Hello, JavaScript World!">
93     </div>
94     <button onclick="runStringMethods()">Run Methods</button>
95
96     <div class="output-box">
97       <strong>Results:</strong>
98       <pre id="methodsOutput">Click "Run Methods" to see the output.</pre>
99     </div>
100   </div>
101
102   <!-- Case Study 1: Reverse String -->
103   <div class="section">
104     <h2>Case Study 1: Reverse a String</h2>
105     <div class="form-group">
106       <label for="reverseInput">Enter text to reverse:</label>
107       <input type="text" id="reverseInput" value="Developer">
108     </div>
109     <button onclick="runReverseString()">Reverse</button>
110
111     <div class="output-box">
112       <strong>Reversed Output:</strong>
113       <pre id="reverseOutput">-</pre>
114     </div>
115   </div>
116
117   <!-- Case Study 2: Count Vowels -->
118   <div class="section">
119     <h2>Case Study 2: Count Vowels in Paragraph</h2>
120     <div class="form-group">
121       <label for="vowelInput">Enter paragraph:</label>
122       <textarea id="vowelInput" rows="4">The quick brown fox jumps over the lazy dog.</textarea>
123     </div>
124     <button onclick="runCountVowels()">Count Vowels</button>
125
126     <div class="output-box">
127       <strong>Vowel Counts:</strong>
128       <pre id="vowelOutput">-</pre>
129     </div>
130   </div>
131 </div>
132
```



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```
151     document.getElementById('methodsOutput').textContent =
152         `Original String: "${str}"\n\n` +
153         `[Basics] Length: ${length}\n` +
154         `[Basics] Character at index 0: "${firstChar}"\n` +
155         `[substring(7, 17)]: "${sub}"\n` +
156         `[indexOf('Java')]: ${idx}\n` +
157         `[split(' ')] : ${JSON.stringify(spl)}\n` +
158         `[replace('World', 'Universe')]: "${rep}"`;
159     }
160
161     // =====
162     // Case Study 1: Reverse a String
163     // =====
164     function reverseString(text) {
165         // 1. split('') breaks the string into an array of characters
166         // 2. reverse() reverses the array in-place
167         // 3. join('') stitches the array elements back into a string
168         return text.split('').reverse().join('');
169     }
170
171     function runReverseString() {
172         const input = document.getElementById('reverseInput').value;
173         const result = reverseString(input);
174         document.getElementById('reverseOutput').textContent = result;
175     }
176
177     // =====
178     // Case Study 2: Count Vowels
179     // =====
180     function countVowels(paragraph) {
181         const vowels = 'aeiouAEIOU';
182         let count = 0;
183         let details = { a: 0, e: 0, i: 0, o: 0, u: 0 };
184
185         // Loop through every character in the string
186         for (let char of paragraph) {
187             if (vowels.includes(char)) {
188                 count++;
189                 let lowerChar = char.toLowerCase();
190                 details[lowerChar] = (details[lowerChar] || 0) + 1;
191             }
192         }
193         return { total: count, details: details };
194     }
195
196     function runCountVowels() {
197         const input = document.getElementById('vowelInput').value;
198         const result = countVowels(input);
199
200         document.getElementById('vowelOutput').textContent =
201             `Total Vowels Found: ${result.total}\n` +
202             `Breakdown: ${JSON.stringify(result.details)}`;
203     }
204 </script>
205
206 </body>
207 </html>
```



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Output:

The screenshot shows a web browser window with the URL `127.0.0.1:5500/JavaScript_Lab/Case_Study/Case%20Study%206/index.html`. The page title is "JavaScript String Operations".

a & b) Basics and Built-in Methods

Input String:

Results:
Click "Run Methods" to see the output.

Case Study 1: Reverse a String

Enter text to reverse:

Reversed Output:
-

Case Study 2: Count Vowels in Paragraph

Enter paragraph:

Vowel Counts:
-

Result:

This experiment demonstrated the effective handling of string data types and manipulation methods in JavaScript. Using built-in methods like `.substring()`, `.split()`, and `.replace()` simplified string searching and parsing without modifying original immutable string variables. Furthermore, combining array techniques and iterative loop checks proved effective for solving practical text processing problems like reversing character sequences and tallying specific target characters.



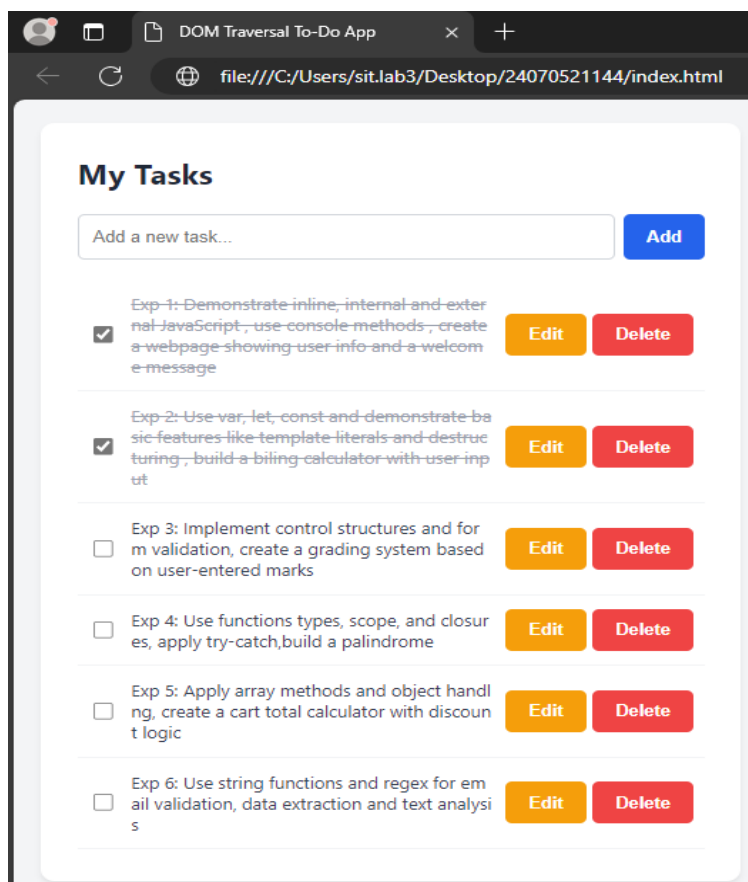
Experiment 07

Aim: Perform DOM traversal and updates, develop a to-do list app with add, edit and delete features.

Theory:

This practical demonstrates client-side DOM (Document Object Model) manipulation, traversal, and event handling by building a dynamic, interactive To-Do list application. It leverages **Event Delegation** by attaching single click and change listeners to the parent `<ul>` container rather than individual items, capturing events as they bubble up from child elements (`event.target`). Upward DOM traversal is performed using `.closest('li')` to identify parent task rows, while downward target selection uses `.querySelector()` to target and mutate descendant nodes. Real-time updates—such as dynamically inserting new elements (`appendChild()`), replacing elements via `.innerHTML` for inline editing, removing elements with `.remove()`, and toggling CSS classes for task completion—are achieved smoothly without requiring server calls or page reloads.

Output:





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Result:

The To-Do application successfully performs real-time client-side task management with full CRUD capability without requiring page reloads. Users can dynamically generate new task items, edit existing task text through an inline input field, delete items instantly from the DOM tree, and toggle completion states via checkboxes to visually strike through finished tasks.

Conclusion:

In conclusion, this project effectively validates the core principles of front-end JavaScript DOM manipulation, traversal, and event handling. By utilizing event delegation on a single parent element and targeting relative nodes using `Element.closest()` and `Element.querySelector()`, the application maintains high performance, clean memory management, and smooth interactive updates for dynamically generated elements.



Case Study 07

Aim: To demonstrate the events handling in javascript

- a) Access and manipulate form elements using js.
- b) Validate user inputs(e.g. Email, empty fields).
- c) Use various form events (submit,change,focus)

Create the following html form and use required js and events in a given form (first name, last name, email, birthday, password, reenter password, agree to the terms and conditions)

Program Code:

```
JavaScript_Lab > Case_Study > Case Study 7 > index.html > html
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Registration Form - Case Study 7</title>
7   <!-- Google Fonts for modern typography -->
8   <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap" rel="stylesheet">
9
10 <style>...
179 </style>
180 </head>
181 <body>
182
183   <div class="form-container">
184     <div class="form-header">
185       <h2>Create an Account</h2>
186       <p>Fill out the form details below to register</p>
187     </div>
188
189     <form id="registrationForm">
190       <div class="form-row">
191         <div class="form-group">
192           <label for="firstName">First Name</label>
193           <input type="text" id="firstName" placeholder="Enter first name">
194           <span class="error" id="firstNameError"></span>
195         </div>
196         <div class="form-group">
197           <label for="lastName">Last Name</label>
198           <input type="text" id="lastName" placeholder="Enter last name">
199           <span class="error" id="lastNameError"></span>
200         </div>
201       </div>
202
203       <div class="form-group">
204         <label for="email">Email</label>
205         <input type="text" id="email" placeholder="example@gmail.com">
206         <span class="error" id="emailError"></span>
207       </div>
208
209       <div class="form-group">
210         <label for="birthday">Birthday</label>
211         <input type="date" id="birthday">
212         <span class="error" id="birthdayError"></span>
213       </div>
214
215       <div class="form-group">
216         <label for="password">Password</label>
217         <input type="password" id="password" placeholder="Enter password">
218         <span class="error" id="passwordError"></span>
219       </div>
220
221       <div class="form-group">
222         <label for="rePassword">Re-enter Password</label>
223         <input type="password" id="rePassword" placeholder="Re-enter password">
224         <span class="error" id="rePasswordError"></span>
225       </div>
226
227       <div class="terms">
228         <input type="checkbox" id="terms">
229         <label for="terms">I agree to the Terms and Conditions</label>
230       </div>
231     </form>
232   </div>
233 </body>
234 </html>
```



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```
229         <label for="terms">I agree to the Terms and Conditions</label>
230     </div>
231     <span class="error" id="termsError"></span>
232
233     <button type="submit">Register</button>
234     <p id="successMessage"></p>
235 </form>
236 </div>
237
238 <script>
239     // Form & Field DOM Access
240     const form = document.getElementById("registrationForm");
241     const firstName = document.getElementById("firstName");
242     const lastName = document.getElementById("lastName");
243     const email = document.getElementById("email");
244     const birthday = document.getElementById("birthday");
245     const password = document.getElementById("password");
246     const rePassword = document.getElementById("rePassword");
247     const terms = document.getElementById("terms");
248
249     // Error Message DOM Access
250     const firstNameError = document.getElementById("firstNameError");
251     const lastNameError = document.getElementById("lastNameError");
252     const emailError = document.getElementById("emailError");
253     const birthdayError = document.getElementById("birthdayError");
254     const passwordError = document.getElementById("passwordError");
255     const rePasswordError = document.getElementById("rePasswordError");
256     const termsError = document.getElementById("termsError");
257     const successMessage = document.getElementById("successMessage");
258
259     // Focus Events (Highlight field background)
260     const textInputs = [firstName, lastName, email, password, rePassword];
261     textInputs.forEach(input => {
262         input.addEventListener("focus", function () {
263             input.style.backgroundColor = "#1e293b";
264         });
265     });
266
267     // Change Events for Live Checks
268     email.addEventListener("change", function () {
269         const emailValue = email.value.trim();
270         const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
271         if (emailValue === "") {
272             emailError.textContent = "Email is required.";
273             emailError.style.color = "#ef4444";
274         } else if (!emailPattern.test(emailValue)) {
275             emailError.textContent = "Please enter a valid email address.";
276             emailError.style.color = "#ef4444";
277         } else {
278             emailError.textContent = "Valid email address.";
279             emailError.style.color = "#10b981";
280         }
281     });
282
283     password.addEventListener("change", function () {
284         if (password.value.length < 6) {
285             passwordError.textContent = "Password must contain at least 6 characters.";
```



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```
313     let isValid = true;
314
315     if (firstName.value.trim() === "") {
316         firstNameError.textContent = "First name is required.";
317         isValid = false;
318     }
319
320     if (lastName.value.trim() === "") {
321         lastNameError.textContent = "Last name is required.";
322         isValid = false;
323     }
324
325     const emailValue = email.value.trim();
326     const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
327     if (emailValue === "") {
328         emailError.textContent = "Email is required.";
329         emailError.style.color = "#ef4444";
330         isValid = false;
331     } else if (!emailPattern.test(emailValue)) {
332         emailError.textContent = "Please enter a valid email address.";
333         emailError.style.color = "#ef4444";
334         isValid = false;
335     }
336
337     if (birthday.value === "") {
338         birthdayError.textContent = "Birthday is required.";
339         isValid = false;
340     }
341
342     if (password.value === "") {
343         passwordError.textContent = "Password is required.";
344         isValid = false;
345     } else if (password.value.length < 6) {
346         passwordError.textContent = "Password must contain at least 6 characters.";
347         isValid = false;
348     }
349
350     if (rePassword.value === "") {
351         rePasswordError.textContent = "Please re-enter your password.";
352         isValid = false;
353     } else if (password.value !== rePassword.value) {
354         rePasswordError.textContent = "Passwords do not match.";
355         isValid = false;
356     }
357
358     if (!terms.checked) {
359         termsError.textContent = "You must agree to the Terms and Conditions.";
360         isValid = false;
361     }
362
363     if (isValid) {
364         successMessage.textContent = "Registration successful!";
365         form.reset();
366     }
367 });
368 </script>
369 </body>
370 </html>
```



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Output:

Registration Form - Case Study

127.0.0.1:5500/JavaScript_Lab/Case_Study/Case%20Study%207/index.html

Hoyolab Tracker Coding Bookmarks LinkedIn Adobe Acrobat

Create an Account

Fill out the form details below to register

FIRST NAME LAST NAME

A S

EMAIL

AS@gmail.com

Valid email address.

BIRTHDAY

01-01-2004

PASSWORD

RE-ENTER PASSWORD

☒ I agree to the Terms and Conditions

Register

Result:

The registration form was successfully created with JavaScript event handling and input validation for email, password, required fields, and terms acceptance. The application dynamically processes user focus and input changes, provides instant error feedback, and confirms completion with a "Registration successful!" message upon valid submission.



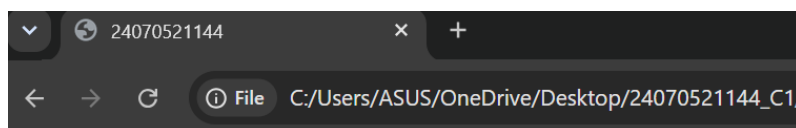
Experiment 08

Aim: Access and validate form fields using events. Design a gym admission form with live input checks.

Theory:

JavaScript event listeners enable interactive, real-time user validation by monitoring form events such as input (triggers on every keystroke) and change (triggers when a selection changes). By attaching listeners to input elements, JavaScript can intercept user inputs immediately, apply regular expression pattern checks or boundary logic, and provide instant visual feedback. Accessing DOM element values directly using `document.getElementById()` allows developers to validate data client-side before form submission. This approach prevents invalid submissions, improves data accuracy, and creates a responsive user experience by giving instant feedback without full-page reloads.

Output:



Gym Admission Form

Full Name

Email Address

Age

Phone Number

Membership Plan

Basic (Monthly) ▾

Submit Admission

Form submitted successfully! Welcome to the Gym.



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Result:

The Gym Admission Form with live input checks was successfully built using HTML, CSS, and JavaScript event handlers. The application continuously monitors user actions across five fields (Full Name, Email Address, Age, Phone Number, and Membership Plan). Real-time validation correctly highlights input borders green for valid entries and red for errors, displays instant contextual error messages, and keeps the submission button disabled until all constraints are met.

Conclusion:

This experiment demonstrated the practical implementation of DOM field access and event-driven form validation in JavaScript. Using real-time event listeners (input and change) proved significantly more user-friendly than traditional form submission checks. Client-side input validation effectively ensures data integrity, improves application responsiveness, and eliminates invalid form submissions prior to server processing.



Case Study 08

Aim: To design and implement a responsive, multi-section Job Application Form styled with modern CSS gradients and card layout.

Program Code:

```
You, 38 minutes ago | author (You)
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Job Application Form</title>
7   <style>...
143 </style>
144 </head>
145 <body>
146
147 <div class="container">
148   <h1>Job Application Form</h1>
149   <p class="subtitle">Please fill in your details accurately. Fields marked with <span class="required">*</span> are mandatory.</p>
150
151   <form id="jobApplicationForm" novalidate>
152     <!-- Personal Information -->
153     <fieldset>
154       <legend>Personal Information</legend>
155       <div class="form-grid">
156         <div class="form-group full-width">
157           <label for="fullName">Full Name <span class="required">*</span></label>
158           <input type="text" id="fullName" name="fullName" placeholder="e.g., John Doe" required>
159           <span class="error-message" id="fullNameError"></span>
160         </div>
161         <div class="form-group">
162           <label for="dob">Date of Birth <span class="required">*</span></label>
163           <input type="date" id="dob" name="dob" required>
164           <span class="error-message" id="dobError"></span>
165         </div>
166         <div class="form-group">
167           <label for="gender">Gender <span class="required">*</span></label>
168           <select id="gender" name="gender" required>
169             <option value="">Select Gender</option>
170             <option value="Male">Male</option>
171             <option value="Female">Female</option>
172             <option value="Other">Other</option>
173             <option value="Prefer not to say">Prefer not to say</option>
174           </select>
175           <span class="error-message" id="genderError"></span>
176         </div>
177         <div class="form-group">
178           <label for="email">Email Address <span class="required">*</span></label>
179           <input type="email" id="email" name="email" placeholder="example@domain.com" required>
180           <span class="error-message" id="emailError"></span>
181         </div>
182         <div class="form-group">
183           <label for="mobile">Mobile Number <span class="required">*</span></label>
184           <input type="tel" id="mobile" name="mobile" placeholder="10-digit number" maxlength="10" required>
185           <span class="error-message" id="mobileError"></span>
186         </div>
187         <div class="form-group full-width">
188           <label for="address">Address <span class="required">*</span></label>
189           <textarea id="address" name="address" placeholder="Street address, apartment, etc." required></textarea>
190           <span class="error-message" id="addressError"></span>
191         </div>
192         <div class="form-group">
193           <label for="city">City <span class="required">*</span></label>
194           <input type="text" id="city" name="city" required>
195           <span class="error-message" id="cityError"></span>
196         </div>
197       </div>
198     </fieldset>
199   </form>
200 </div>
201 </body>
202 </html>
```



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```
209 <!-- Educational Qualification -->
210 <fieldset>
211 <legend>Educational Qualification</legend>
212 <div class="form-grid">
213 <div class="form-group">
214 <label for="qualification">Highest Qualification <span class="required">*</span></label>
215 <select id="qualification" name="qualification" required>
216 <option value="">Select Qualification</option>
217 <option value="High School">High School</option>
218 <option value="Diploma">Diploma</option>
219 <option value="Bachelor's">Bachelor's Degree</option>
220 <option value="Master's">Master's Degree</option>
221 <option value="Ph.D.">Ph.D.</option>
222 </select>
223 <span class="error-message" id="qualificationError"></span>
224 </div>
225 <div class="form-group">
226 <label for="university">University / Institute <span class="required">*</span></label>
227 <input type="text" id="university" name="university" required>
228 <span class="error-message" id="universityError"></span>
229 </div>
230 <div class="form-group">
231 <label for="passingYear">Passing Year <span class="required">*</span></label>
232 <input type="number" id="passingYear" name="passingYear" placeholder="e.g., 2023" min="1980" max="2030" required>
233 <span class="error-message" id="passingYearError"></span>
234 </div>
235 <div class="form-group">
236 <label for="cgpa">CGPA / Percentage <span class="required">*</span></label>
237 <input type="number" id="cgpa" name="cgpa" placeholder="e.g., 8.5" step="0.01" min="0" max="100" required>
238 <span class="error-message" id="cgpaError"></span>
239 </div>
240 </div>
241 </fieldset>
242
243 <!-- Professional Information -->
244 <fieldset>
245 <legend>Professional Information</legend>
246 <div class="form-grid">
247 <div class="form-group">
248 <label for="jobPosition">Applying For / Job Position <span class="required">*</span></label>
249 <input type="text" id="jobPosition" name="jobPosition" placeholder="e.g., Software Engineer" required>
250 <span class="error-message" id="jobPositionError"></span>
251 </div>
252 <div class="form-group">
253 <label for="experience">Total Work Experience (Years) <span class="required">*</span></label>
254 <input type="number" id="experience" name="experience" placeholder="e.g., 3" min="0" max="50" required>
255 <span class="error-message" id="experienceError"></span>
256 </div>
257 <div class="form-group full-width">
258 <label for="organization">Current / Previous Organization <span class="required">*</span></label>
259 <input type="text" id="organization" name="organization" required>
260 <span class="error-message" id="organizationError"></span>
261 </div>
262 <div class="form-group">
263 <label for="expectedSalary">Expected Salary (Annual) <span class="required">*</span></label>
264
```



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```
293 <script>
294 // Utility function to show error
295 function showError(inputId, message) {
296     const input = document.getElementById(inputId);
297     const errorSpan = document.getElementById(inputId + 'Error');
298     input.classList.add('invalid');
299     input.classList.remove('valid');
300     if (errorSpan) errorSpan.textContent = message;
301 }
302
303 // Utility function to clear error / show success
304 function clearError(inputId) {
305     const input = document.getElementById(inputId);
306     const errorSpan = document.getElementById(inputId + 'Error');
307     input.classList.remove('invalid');
308     input.classList.add('valid');
309     if (errorSpan) errorSpan.textContent = '';
310 }
311
312 // Validation Rules
313 const validators = {
314     fullName: (val) => {
315         if (!val.trim()) return "Full name is required.";
316         if (val.trim().length < 3) return "Name must be at least 3 characters.";
317         if (!/^[a-zA-Z\s]+$/.test(val)) return "Name should only contain letters.";
318         return "";
319     },
320     dob: (val) => {
321         if (!val) return "Date of birth is required.";
322         const dob = new Date(val);
323         const today = new Date();
324         let age = today.getFullYear() - dob.getFullYear();
325         const m = today.getMonth() - dob.getMonth();
326         if (m < 0 || (m === 0 && today.getDate() < dob.getDate())) age--;
327         if (age < 18) return "You must be at least 18 years old.";
328         if (dob > today) return "Date of birth cannot be in the future.";
329         return "";
330     },
331     gender: (val) => !val ? "Please select a gender." : "",
332     email: (val) => {
333         if (!val) return "Email is required.";
334         const regex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
335         return !regex.test(val) ? "Please enter a valid email address." : "";
336     },
337     mobile: (val) => {
338         if (!val) return "Mobile number is required.";
339         return !/^[0-9]{10}$/.test(val) ? "Enter a valid 10-digit mobile number." : "";
340     },
341     address: (val) => !val.trim() ? "Address is required." : "",
342     city: (val) => !val.trim() ? "City is required." : "",
343     state: (val) => !val.trim() ? "State is required." : "",
344     pincode: (val) => {
345         if (!val) return "PIN code is required.";
346         return !/^[0-9]{6}$/.test(val) ? "Enter a valid 6-digit PIN code." : "";
347     },
348     qualification: (val) => !val ? "Please select your highest qualification." : "",
349     university: (val) => !val.trim() ? "University name is required." : "",
350     passingYear: (val) => {
```



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```
385         const errorMsg = validators[id](element.value);
386         if (errorMsg) {
387             showError(id, errorMsg);
388         } else {
389             clearError(id);
390         }
391     });
392 });
393 }
394 });
395
396 // Form Submission Handling
397 document.getElementById('jobApplicationForm').addEventListener('submit', function(e) {
398     e.preventDefault(); // Prevent default submission
399
400     let isValid = true;
401     let firstInvalidField = null;
402
403     // Run all validators
404     Object.keys(validators).forEach(id => {
405         const element = document.getElementById(id);
406         const errorMsg = validators[id](element.value);
407
408         if (errorMsg) {
409             showError(id, errorMsg);
410             isValid = false;
411             if (!firstInvalidField) firstInvalidField = element;
412         } else {
413             clearError(id);
414         }
415     });
416
417     if (!isValid) {
418         // Scroll to the first error
419         firstInvalidField.scrollIntoView({ behavior: 'smooth', block: 'center' });
420         firstInvalidField.focus();
421         return;
422     }
423
424     // If valid, simulate submission
425     const submitBtn = document.getElementById('submitBtn');
426     submitBtn.disabled = true;
427     submitBtn.textContent = "Submitting...";
428
429     setTimeout(() => {
430         document.getElementById('successMsg').style.display = 'block';
431         submitBtn.textContent = "Submitted";
432         // In a real scenario, you would send the data to a server here using fetch() or AJAX
433         console.log("Form Data Submitted:", new FormData(this));
434     }, 1000);
435 });
436 </script>
437
438 </body>
439 </html>
```



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Output:

Job Application Form

Please fill in your details accurately. Fields marked with * are mandatory.

Personal Information

Full Name *

e.g., John Doe

Date of Birth *

dd-mm-yyyy

Gender *

Select Gender

Email Address *

example@domain.com

Mobile Number *

10-digit number

Address *

Street address, apartment, etc.

City *

State *

PIN Code *

6-digit PIN

Educational Qualification

Highest Qualification *

Select Qualification

University / Institute *

Passing Year *

e.g., 2023

CGPA / Percentage *

e.g., 8.5

Professional Information



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Educational Qualification

Highest Qualification *

University / Institute *

Passing Year *

CGPA / Percentage *

Professional Information

Applying For / Job Position *

Total Work Experience (Years) *

Current / Previous Organization *

Expected Salary (Annual) *

Technical Skills

Programming Skills *

Certifications

Submit Application

Result: This experiment demonstrated the end-to-end design and implementation of client-side form validation for a complex multi-section application form. Centralizing validation logic inside a structured JavaScript validators lookup object streamlined real-time input checks and DOM manipulation. Utilizing event-driven validation improves user feedback, preserves data integrity prior to form submission, and ensures a responsive user experience.



Experiment 09

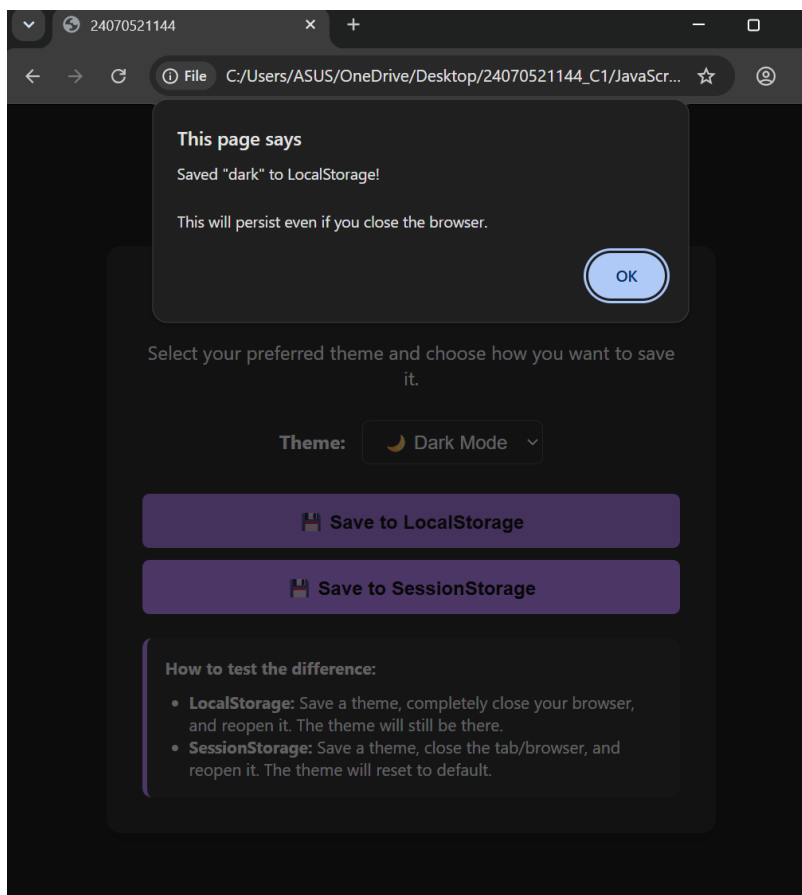
Aim: Implement Local and Session storage to save user preferences like theme settings

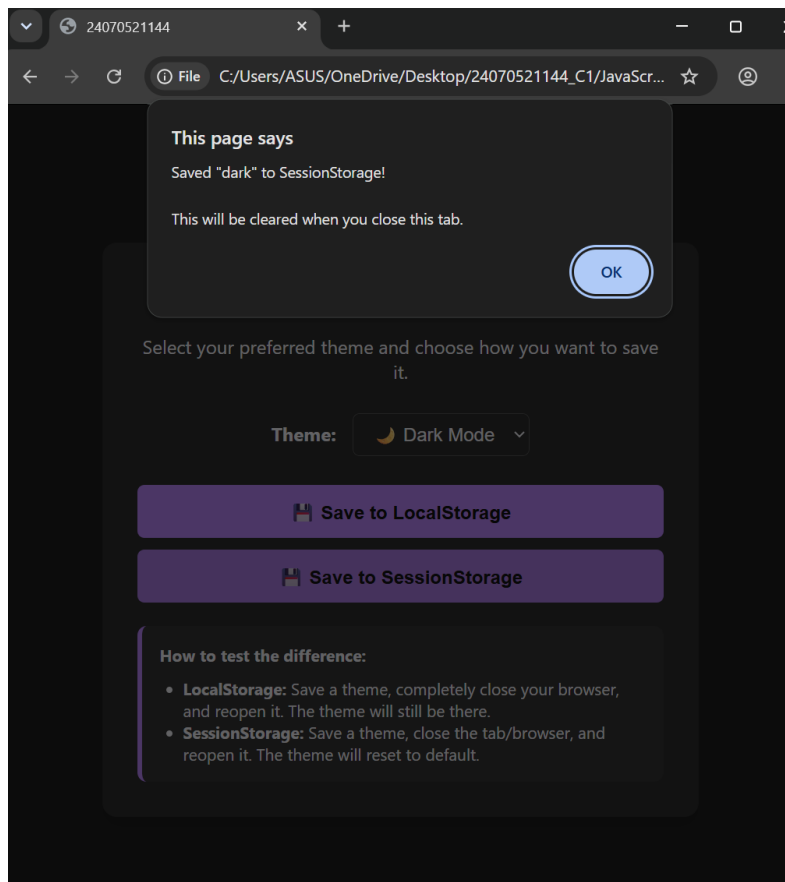
Theory:

Web Storage API provides key-value storage mechanisms—localStorage and sessionStorage—that allow web applications to store data locally within the user's browser without sending it to a server. localStorage persists data permanently across browser sessions until explicitly cleared, making it ideal for storing long-term user preferences such as theme settings (dark/light mode).

sessionStorage, on the other hand, maintains data only for the duration of the current page session and clears automatically when the browser tab or window closes. Data in Web Storage is stored as strings, requiring getItem(), setItem(), and removeItem() methods for manipulation. In this experiment, these storage mechanisms are implemented to save, retrieve, and automatically apply a user's selected theme preferences across page reloads and browser sessions.

Output:





Result: The theme preference manager was successfully implemented using localStorage and sessionStorage. The application accurately captures user input (such as toggling between light and dark themes), saves the selected state in the browser's web storage, and applies the preferred theme instantly via DOM manipulation. Upon reloading the page or reopening the tab, the application successfully retrieves the saved preference from storage and maintains the correct theme without manual reselection.

Conclusion: This experiment demonstrated the practical usage of the Web Storage API for managing client-side state and user preferences. Utilizing localStorage proved effective for maintaining persistent application states like theme modes across browser restarts, while sessionStorage demonstrated its utility for tab-bound temporary data. Implementing client-side storage reduces unnecessary server requests, speeds up initial page styling on load, and enhances overall user experience through persistent personalization.`



Experiment 10

Aim: Load and display JSON data using `fetch()` and `jQuery$.getJSON()`, render results dynamically in a table

Theory:

The JavaScript Fetch API and jQuery library provide robust asynchronous methods for retrieving and rendering structured data (such as JSON) dynamically on web pages without triggering a full page reload. Local JSON arrays allow client-side modeling of structured records, which can be queried asynchronously using JavaScript Promises or jQuery helper methods (`$.each()`). When data is fetched, DOM manipulation techniques iterate through the array to generate dynamic HTML table elements (`<tr>` and `<td>`), replacing static placeholders. Furthermore, integrating user feedback mechanisms—such as dynamic status text and loading indicators—ensures an asynchronous user experience by signaling data processing states during execution.

Output:

The screenshot shows a web browser window with the URL `127.0.0.1:5500/JavaScript_Lab/Experiments/Exp10/index.html`. The page displays a "Team Members" section with the subtitle "Fetch and render local JSON dataset using Fetch API or jQuery". There are two buttons: "Load via Fetch Logic" (blue) and "Load via jQuery Logic" (orange). Below the buttons, a status message says "Data loaded successfully.". A table with 3 columns (ID, NAME, EMAIL) displays the following data:

ID	NAME	EMAIL
24070521144	Atharva	atharva.suryawanshi.batch2024@sitnagpur.siu.edu.in
24070521146	Soumya	soumya.bajirao.batch2024@sitnagpur.siu.edu.in
24070521211	Janhavi	janhavi.admane.batch2024@sitnagpur.siu.edu.in
24070521143	Arman	arman.meshram.batch2024@sitnagpur.siu.edu.in



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Result: The Local JSON Data Table application was successfully built and tested using both native JavaScript Fetch logic (simulated using Promises) and jQuery array iteration methods. Upon clicking either the "Load via Fetch Logic" or "Load via jQuery Logic" button, the application triggers a loading animation and updates the status message to indicate processing. After an artificial delay simulating a network request, the local JSON team dataset—containing member ID numbers, names, and official institutional email addresses—is dynamically parsed and injected into the target HTML table. Both methods consistently generated identical table structures, updated the status text to confirm successful data loading, and hid the loader once rendering was complete.

Conclusion: This experiment demonstrated the practical implementation of client-side data rendering using JavaScript Promises and jQuery. Simulating asynchronous retrieval highlighted how modern Web APIs and JS libraries process structured JSON data to dynamically update DOM elements without refreshing the web page. Both the Fetch API logic and jQuery methods proved effective for handling local data arrays, ensuring efficient data presentation and a seamless, responsive user interface.